

Library Management System Using JDBC (Complete Project)

Project Structure:
LibraryManagementStatement
 ■■■ src
 ■■■■ Book.java
 ■■■■ BookDAO.java
 ■■■■ Main.java

Database Setup:
CREATE DATABASE librarydb;
USE librarydb;

```
CREATE TABLE books (  
    book_id INT PRIMARY KEY,  
    title VARCHAR(50),  
    author VARCHAR(50),  
    category VARCHAR(30),  
    price DOUBLE  
);
```

Book.java

```
public class Book {  
    private int bookId;  
    private String title;  
    private String author;  
    private String category;  
    private double price;  
  
    public Book(int bookId, String title, String author, String category, double price) {  
        this.bookId = bookId;  
        this.title = title;  
        this.author = author;  
        this.category = category;  
        this.price = price;  
    }  
  
    public int getBookId(){ return bookId; }  
    public String getTitle(){ return title; }  
    public String getAuthor(){ return author; }  
    public String getCategory(){ return category; }  
    public double getPrice(){ return price; }  
}
```

BookDAO.java

```
import java.sql.*;  
  
public class BookDAO {  
    Connection con;  
  
    public BookDAO() throws Exception {  
        Class.forName("com.mysql.cj.jdbc.Driver");  
        con = DriverManager.getConnection(  
            "jdbc:mysql://localhost:3306/librarydb?useSSL=false&serverTimezone=UTC",  
            "root",  
            "your_password"  
        );  
    }  
  
    public void insertBook(Book b) throws Exception {  
        Statement st = con.createStatement();  
        String sql = "insert into books values(" +  
            b.getBookId() + ", '" +
```



```

        double price = sc.nextDouble();

        Book b = new Book(id, title, author, category, price);
        dao.insertBook(b);
        System.out.println("Book Inserted Successfully!");
        break;

    case 2:
        System.out.println("Enter Book ID:");
        int uid = sc.nextInt();

        System.out.println("Enter New Price:");
        double newPrice = sc.nextDouble();

        dao.updatePrice(uid, newPrice);
        System.out.println("Price Updated!");
        break;

    case 3:
        System.out.println("Enter Book ID:");
        int did = sc.nextInt();

        dao.deleteBook(did);
        System.out.println("Book Deleted!");
        break;

    case 4:
        sc.nextLine();
        System.out.println("Enter Category:");
        String cat = sc.nextLine();

        dao.displayCategory(cat);
        break;

    case 5:
        System.exit(0);
    }
}
}
}
}
}

```